

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_11apr25_isolado.parquet

model_type: LSTM
group_sec: 80
timesteps_orig: 60
batch_size: 256
min_units: 16
max_units: 256
units_step: 16
dropout_min: 0.2
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 20
hyperband_iterations: 3
group_msec: 80
samples_per_sec: 12
forecast_horizon_sec: 4
forecast_steps: 48
timesteps: 60

LSTM Multi-Step Training Report

Report Generated: 2025-08-01 14:28:18

Training Start: 2025-08-01_14-18-17

Training End: 2025-08-01_14-25-57

Training Duration: 7.67 minutes

Predictions Start: 2025-08-01_14-25-57

Predictions End: 2025-08-01_14-26-53

Predictions Duration: 0.92 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: tanh

Optimizer: Nadam

Dropout Rate: 0.20

MAE: 1.9633

MSE: 6.9293

RMSE: 2.6324

R2 Score: -0.0649

Multi-Step Forecast:

Forecast Steps: 48

MAE: 0.0501

MSE: 0.0046

RMSE: 0.0679

R2 Score: 0.5103

Dataset size before down-sample: 40,318 rows
Dataset size after down-sample: 6,059 rows

First 5 timestamps BEFORE:

1744397593300
1744397593310
1744397593332
1744397593362
1744397593375

...

Last 5 timestamps BEFORE:

1744398077967
1744398077978
1744398077991
1744398078003
1744398078016

First 5 timestamps AFTER:

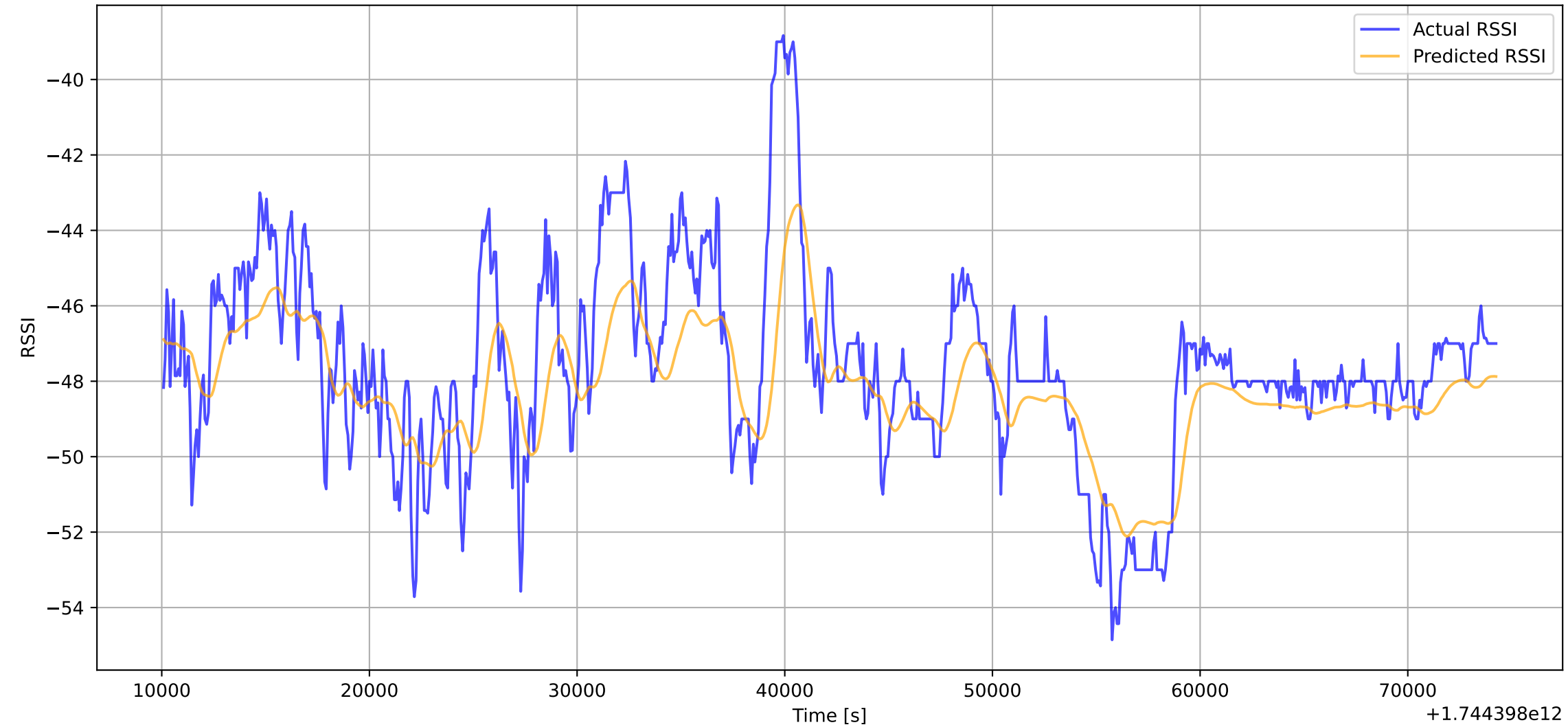
1744397593300
1744397593362
1744397593536
1744397593601
1744397593696

...

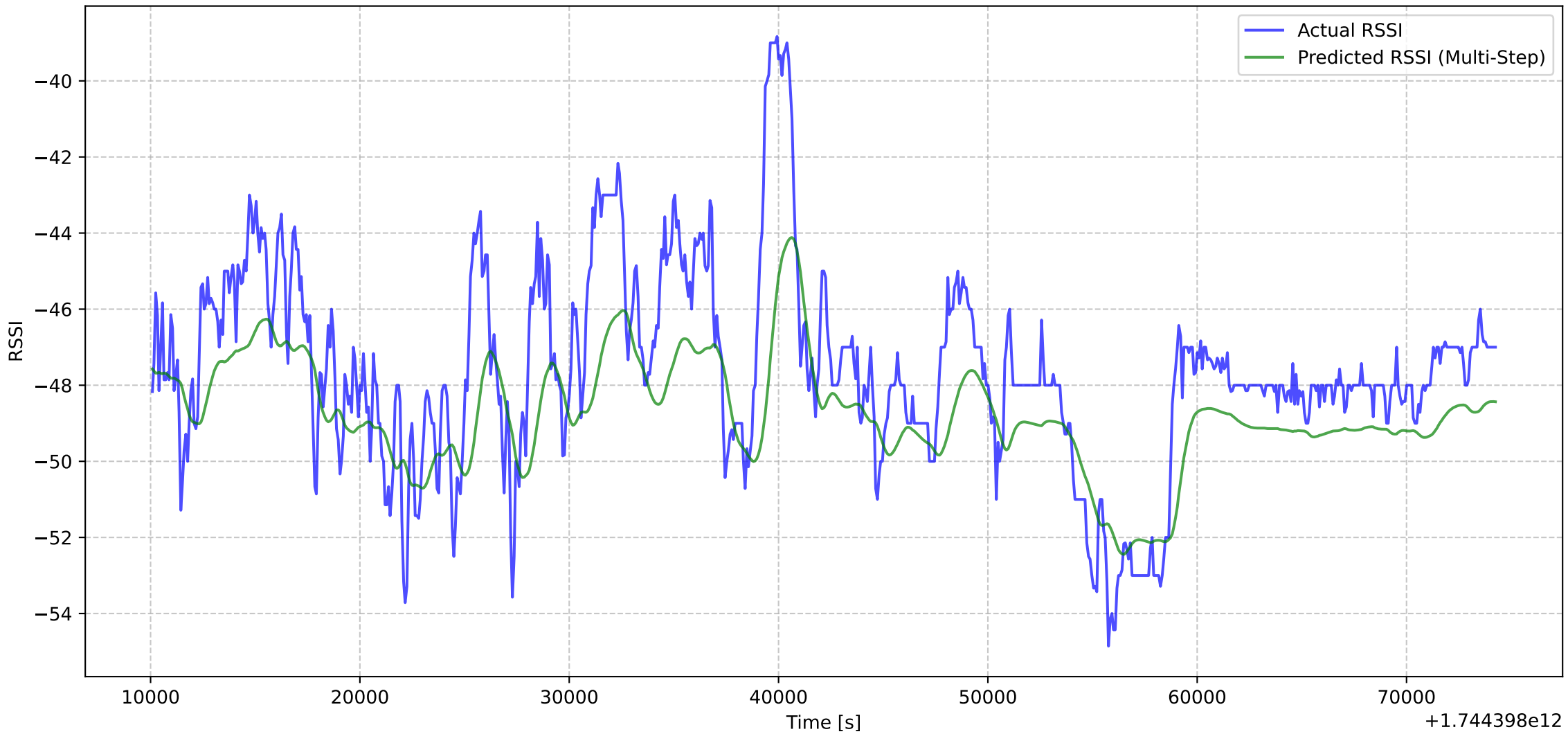
Last 5 timestamps AFTER:

1744398077680
1744398077763
1744398077847
1744398077921
1744398078003

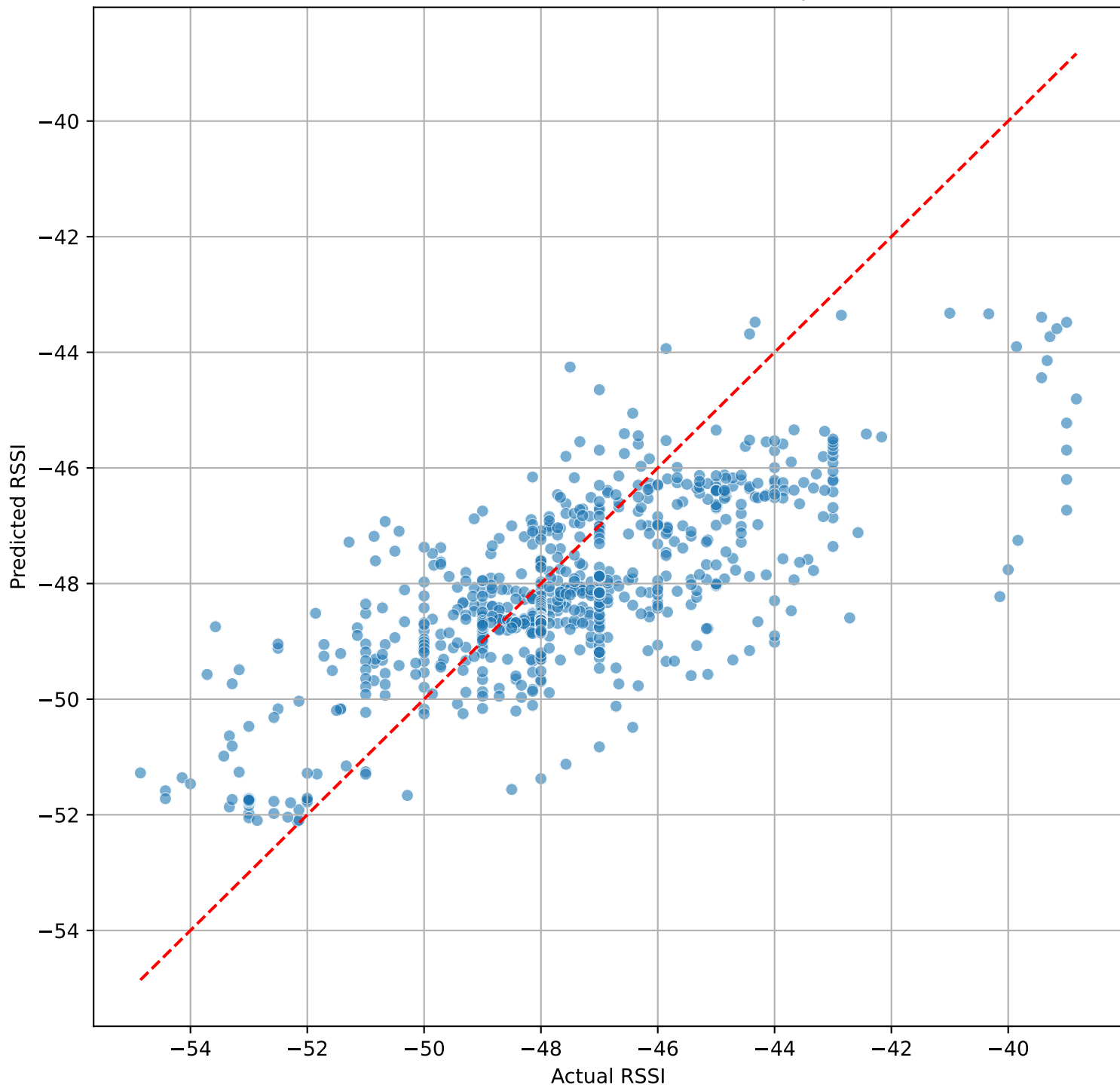
Actual vs Predicted RSSI (First Step)



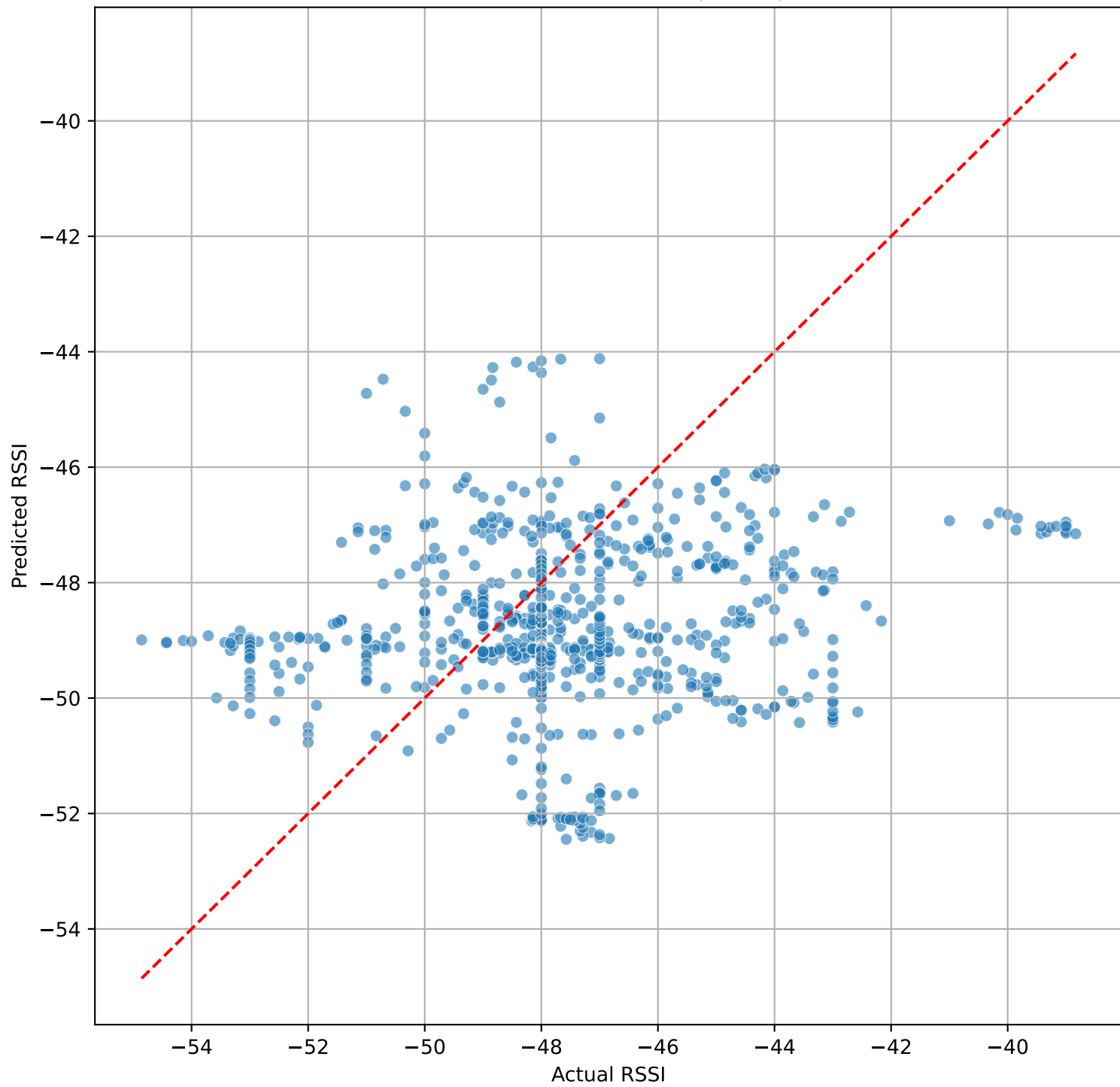
Actual vs Predicted RSSI (Multi-Step (step 48 $\approx$ 4.0 s))



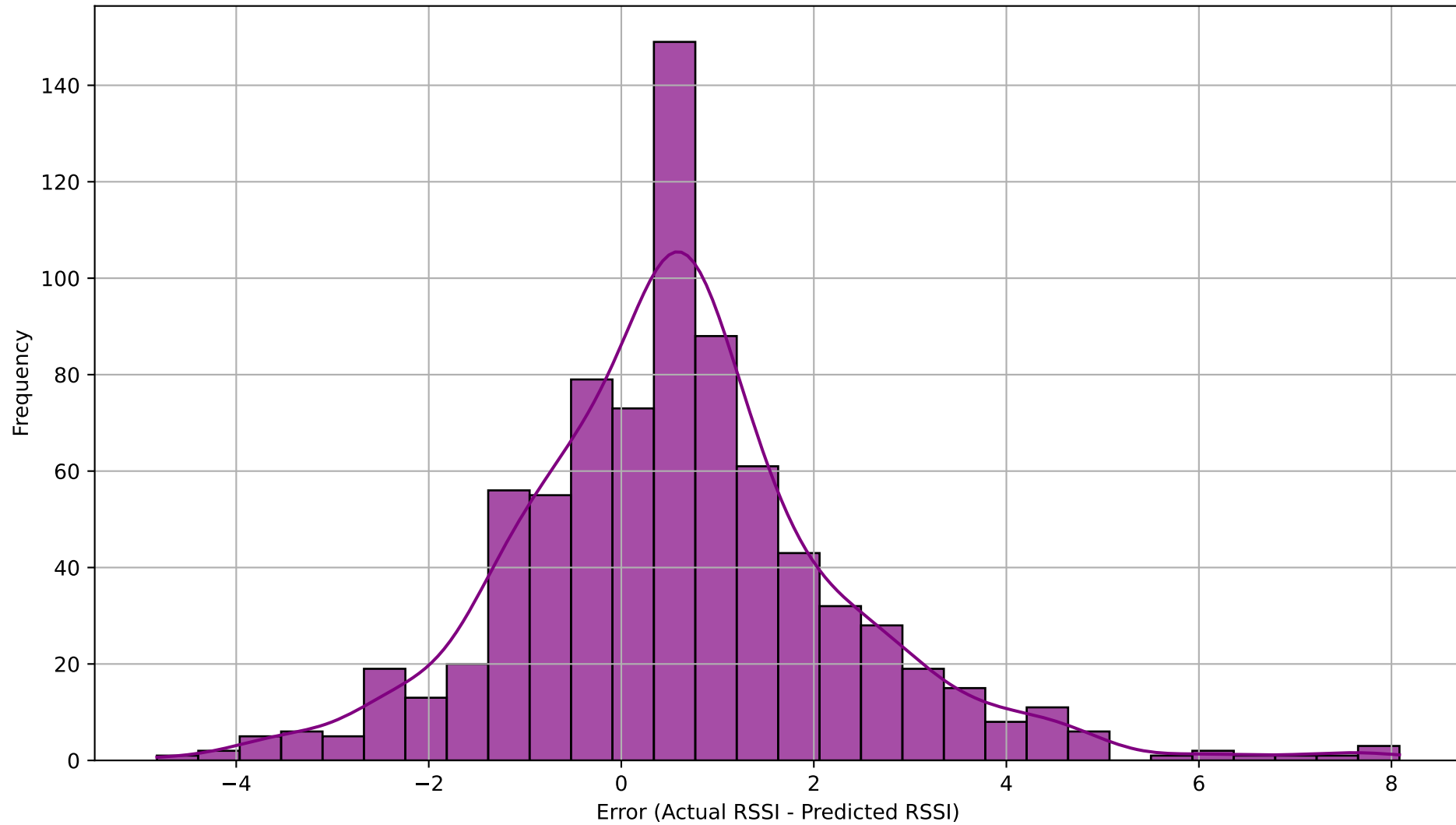
Actual vs Predicted RSSI (First Step)



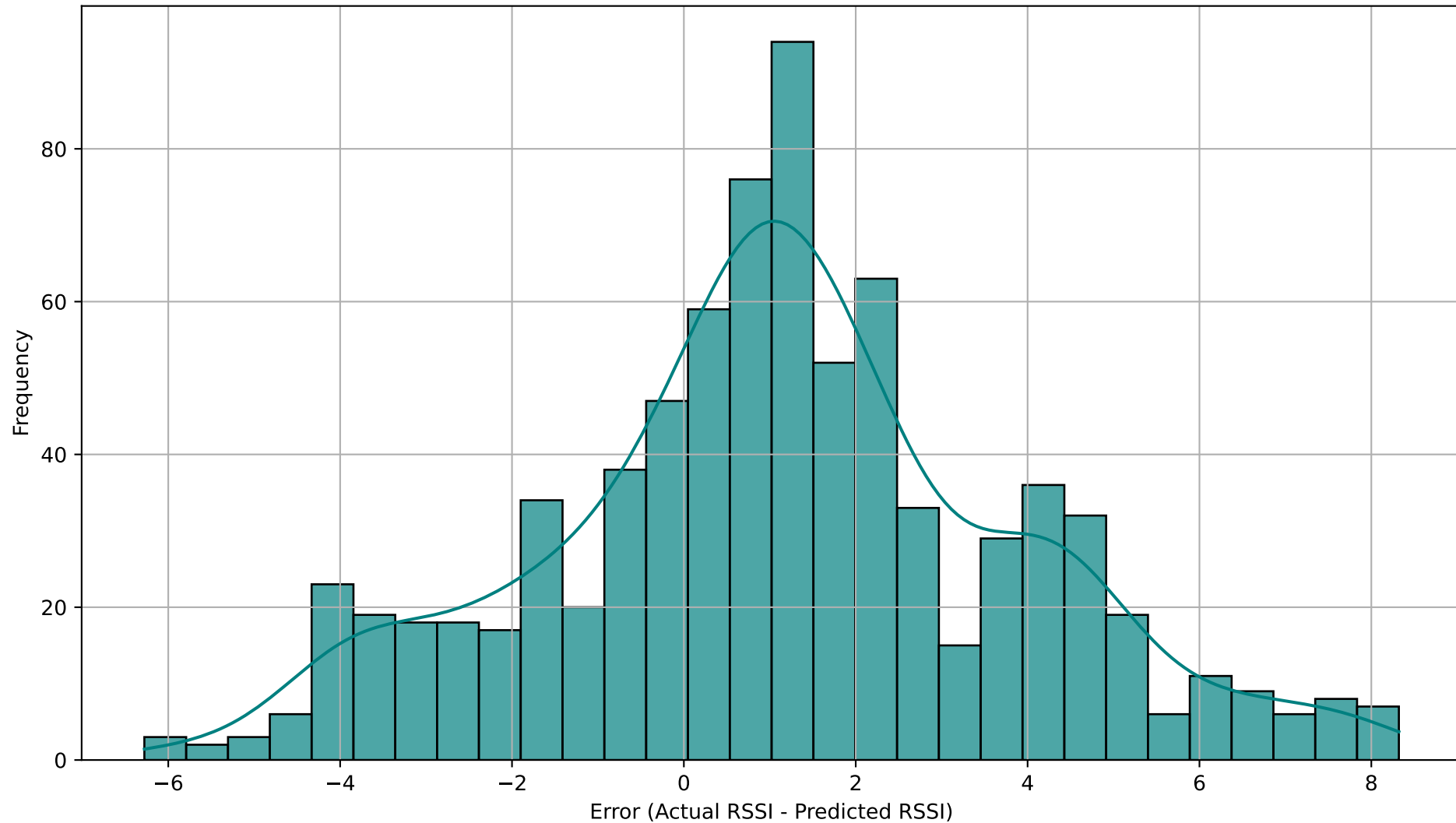
Actual vs Predicted RSSI (Last Step - step 48 $\approx$ 4.0 s)



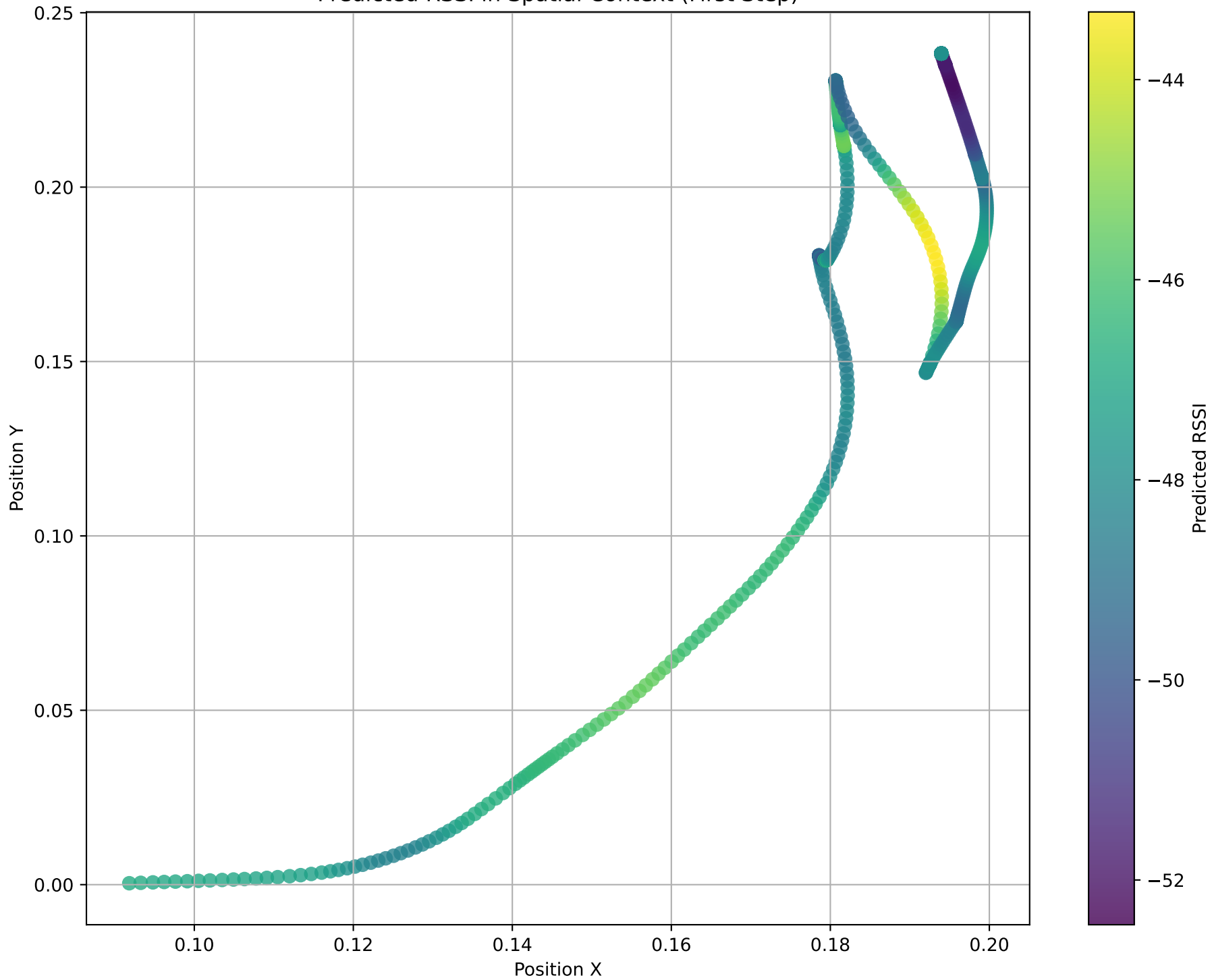
Distribution of Prediction Errors (First Step)



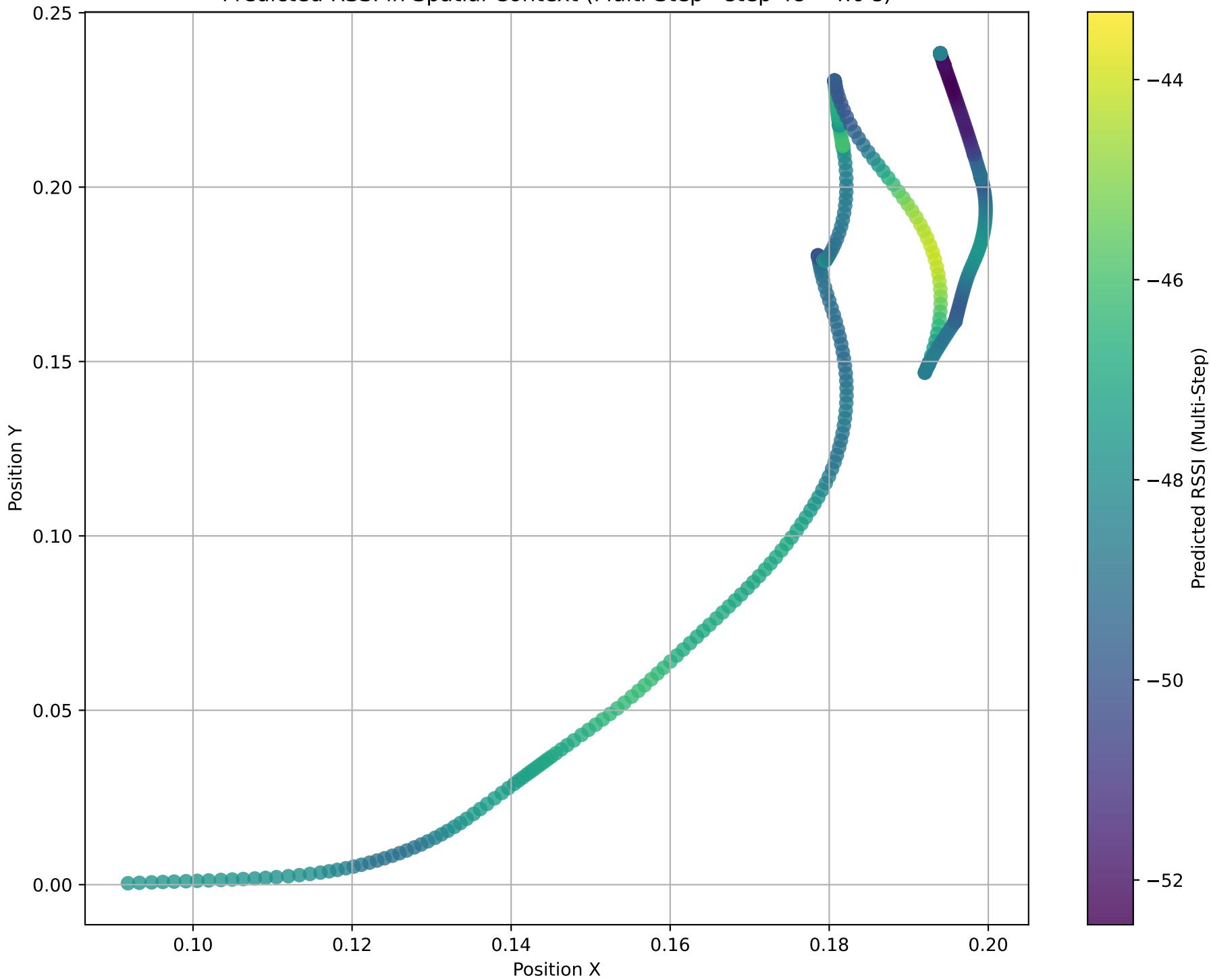
Distribution of Prediction Errors (Last Step - step 48 $\approx$ 4.0 s)



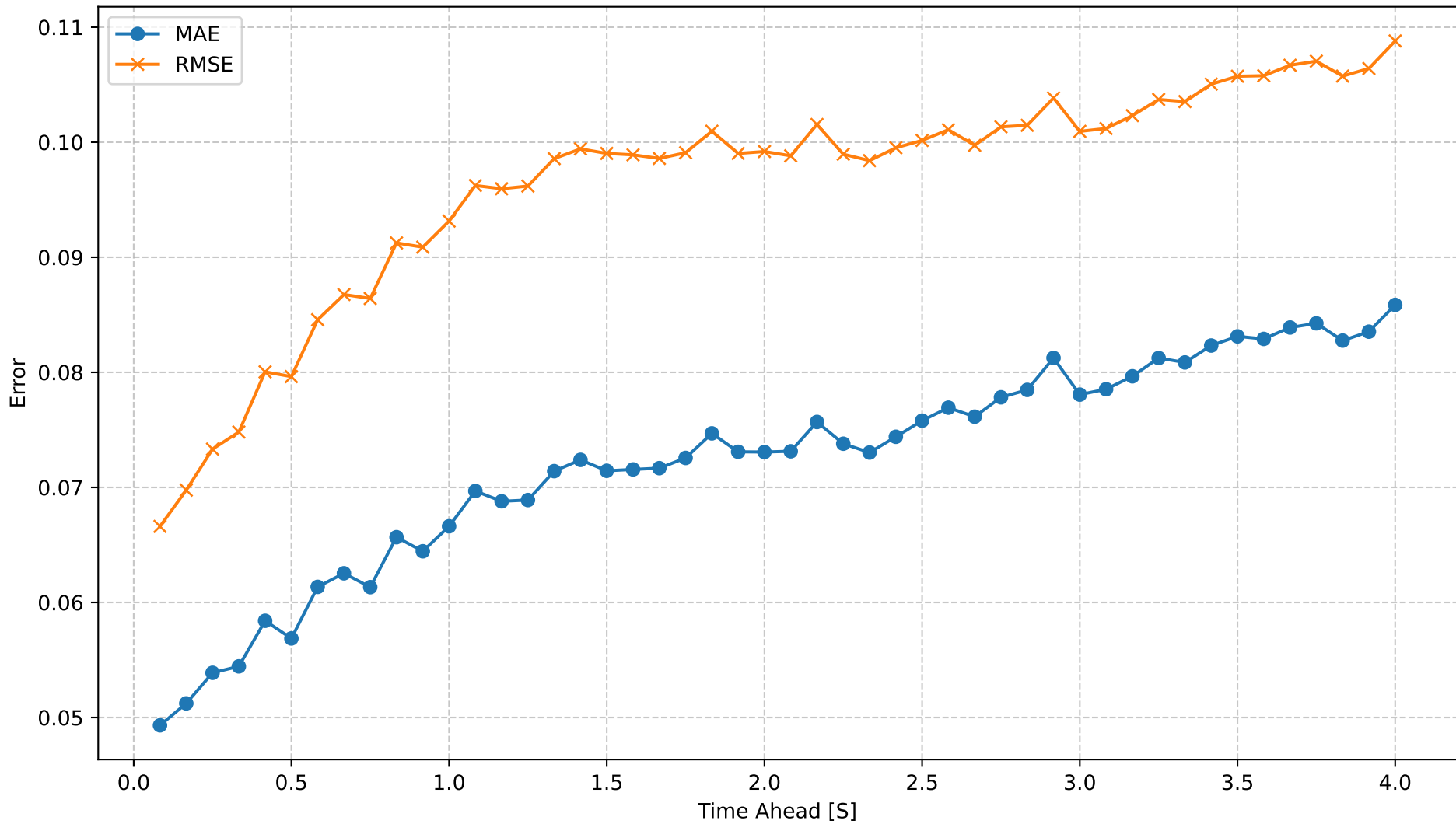
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 48 $\approx$ 4.0 s)



Multi-Step Forecast Error by Time Ahead



Resumo Automático

Dataset: rssi_odom_dataset_11apr25_isolado.parquet
Modelo: LSTM
Forecast steps: 48

MAE / RMSE (dB)

- 1º passo : 0.05 / 0.07
- Mediana : 0.07 / 0.10
- 48º passo : 0.09 / 0.11

Crescimento do erro: 74.1% → Degrada Rápido (> 50 %)

3 piores passos (MAE): 48 (0.09), 45 (0.08), 44 (0.08)

Recomendação:

Considere aumentar o histórico de entrada ou usar um modelo mais profundo para melhorar multi-step.